

$$\{x^m, \chi^m\} \quad m=0, \dots, D-1$$

$$x^m: \mathbb{M}^D$$

$$\chi_A: \text{Majorana}$$

$$\chi_A = \begin{pmatrix} \psi_\alpha \\ \psi^\dagger_{\dot{\alpha}} \end{pmatrix}, (\chi_A)^C = \chi_A^C$$

$$\chi = C \bar{\chi}^T$$

$$\text{or } \bar{\chi} = \chi^T C$$

$\leadsto$ Spiner on Σ but not in $\mathbb{M}^D$

$$(1,1) \text{ Susy: } (Q_\alpha, \bar{Q}_{\dot{\alpha}})$$

$$\bullet \text{ Supermultiplet } \{x^m, \chi_A^m, N^m\}$$

$\leadsto$ Take scalar multiplet $\{\phi, \psi, F\} \rightarrow$ to the vector rep of $\mathbb{M}^D$

$$\mu: \mathbb{M}^D$$

$$A: \text{Spin}(2) \text{ on } \Sigma_2$$

$$\alpha: \Sigma_2 \cong \mathbb{M}^2 \text{ (via gauge fixing)}$$

$$\int d^2x \left(-\frac{1}{2} \partial_\alpha x^m \partial^\alpha x^m - \frac{i}{2} \bar{\chi}^m \not{\partial} \chi^m + \frac{1}{2} N^m N^m \right) \mathcal{H}_{mN}$$

(1,1) rigid susy

* The super charges are added to create a Majorana super charge $\hat{Q}_A$

$$\leadsto (1,1) \text{ Susy: } Q_A^i, \quad i=1,2$$

$$\hat{Q}_A = \frac{1}{2} (Q_A^1 - i Q_A^2)$$

$$\bar{\hat{Q}}^A = \frac{1}{2} (\bar{Q}^{1A} + i \bar{Q}^{2A})$$

$$\{Q_A^i, \bar{Q}^{jB}\} = -i \delta^{ij} (\gamma^\delta)_A^B P_\delta$$

$$\delta \phi = i \bar{\epsilon} \chi, \quad \delta \chi = (\not{\partial} \phi + N) \epsilon, \quad \delta N = i \bar{\epsilon} \not{\partial} \chi$$

* Coupling this field to supergravity mult. $(e_\alpha^a, \psi_{\alpha A})$

$\leadsto$ Use Noether procedure to couple

$$(\phi, \psi, N) \rightarrow (x^m, \chi_A^m, N^m) \oplus (e_\alpha^a, \psi_{\alpha A})$$

From this we obtain the Action

$$S = \frac{1}{2\kappa^2} \int d^2x e \left[-\frac{1}{2} e_\alpha^a \partial_\alpha x^m \eta^{ab} e_b^\beta \partial_\beta x^m - \frac{i}{2} \bar{\chi}^m \not{\partial} \chi^m + \frac{1}{2} N^m N^m + \frac{i\kappa}{2} \bar{\psi}_A \not{\partial}_\mu \chi^m \gamma^\mu \gamma^5 \psi^A - \frac{\kappa^2}{2} \bar{\psi}_A \gamma^\mu \gamma^5 \psi^A \partial_\mu x^m \right]$$

$\times \eta_{\mu\nu}$